Causally-Constrained Counterfactual Explanations for GC-MS Retention Time Prediction

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Abstract. This paper presents a four-step pipeline that extends an XGBoost-based GC-MS retention time prediction model with causally-constrained counterfactual explanations. Counterfactuals are generated using DiCE (Diverse Counterfactual Explanations) [1] and validated against a DoWhy causal graph [2] encoding domain knowledge from analytical chemistry — a directed acyclic graph (DAG) that maps molecular structural features through physicochemical mediators to retention time [3]. Evaluation across 16 molecules (80 counterfactuals) demonstrates a **95% causal pass rate**, rising to **100%** after constraint enforcement, with no systematic proximity cost ($r = -0.14$, $p = 0.603$). The pipeline produces physically interpretable, causally grounded explanations across the full GC-MS retention time range (7–38 min).

1. Introduction

GC-MS retention time prediction models based on machine learning achieve high predictive accuracy but remain black boxes — a chemist cannot readily understand why a given molecule elutes at a particular time. Explainable AI (XAI) addresses this gap by making model decisions interpretable [4]. Counterfactual explanations are a particularly useful XAI technique: they answer the question "what would need to change about this molecule for it to elute at a different time?" — an approach pioneered in the XAI context by Wachter et al. [5].

Standard counterfactual methods such as DiCE [1] optimise for statistical validity and diversity, but have no concept of physical or causal plausibility. A counterfactual that suggests increasing a downstream descriptor while decreasing its upstream cause is chemically impossible, even if it satisfies the model. This paper introduces a causal constraint layer using DoWhy [2] that enforces the causal ordering of molecular descriptors as described by Pearl's framework [3, 6], ensuring every explanation is physically interpretable.

The pipeline extends the CS 6610 capstone model (XGBoost [7], $R^2 = 0.9714$, 37 RDKit [8] molecular features, 2,000 molecules) with three additional components: DiCE counterfactual generation [1], DoWhy causal validation [2], and a comprehensive evaluation framework. Recent

work on GC-MS retention time prediction using ensemble ML methods confirms the effectiveness of XGBoost with RDKit descriptors for chromatographic property prediction [9, 10].

2. Methods

2.1 Causal Graph

The foundation of the constraint layer is a hand-crafted DAG encoding analytical chemistry domain knowledge, following the structural causal model framework of Pearl [3]. The graph has four layers:

1. **Root causes:** Molecular Weight, Aromatic Rings, Fraction CSP^3, Heteroatoms — primary structural variables determined directly by molecular structure (SMILES).
2. **Mediators:** LogP, TPSA, MolMR, Labute ASA, Chi0v, Kappa1 — physicochemical properties causally derived from root causes and computed via RDKit [8].
3. **Derived features:** PEOE_VSA 1–5, SLogP_VSA 1–2, SMR_VSA 1–2, H-Acceptors/H-Donors, Kappa2, BalabanJ — downstream computed descriptors.
4. **Outcome:** Retention Time — the target variable predicted by the XGBoost model [7].

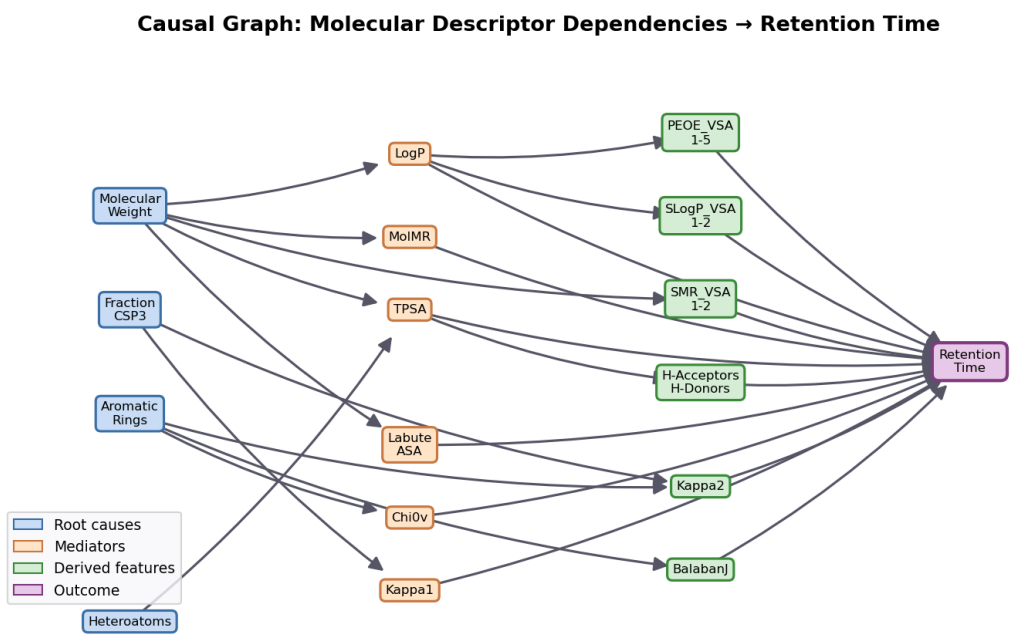

Figure 1: Causal graph of molecular descriptor dependencies.

2.2 Pipeline Steps

1. **Step 1:** XGBoost [7] model predicts retention time for the query molecule.
2. **Step 2:** DiCE [1] generates 5 diverse counterfactuals targeting a ±2 min RT window.

3. **Step 3:** DoWhy [2] validates each counterfactual against the causal graph; violations are flagged and replaced.
4. **Step 4:** Metrics are computed and visualisations generated.

3. Results

3.1 Experiment 1: Causal Constraint Violation Analysis

Across 80 counterfactuals generated for 16 molecules, the DoWhy [2] filter achieved a **95% causal pass rate** — 76 of 80 counterfactuals were causally valid without intervention. The 4 violations were entirely concentrated in a single molecule, triethylamine (`CCN(CC)CC`), whose branched tertiary amine creates descriptor relationships that DiCE's [1] unconstrained search violated. After constraint enforcement, 16/16 molecules achieved **100% causal validity**.

Figure 2. Experiment 1: Causal constraint violation analysis

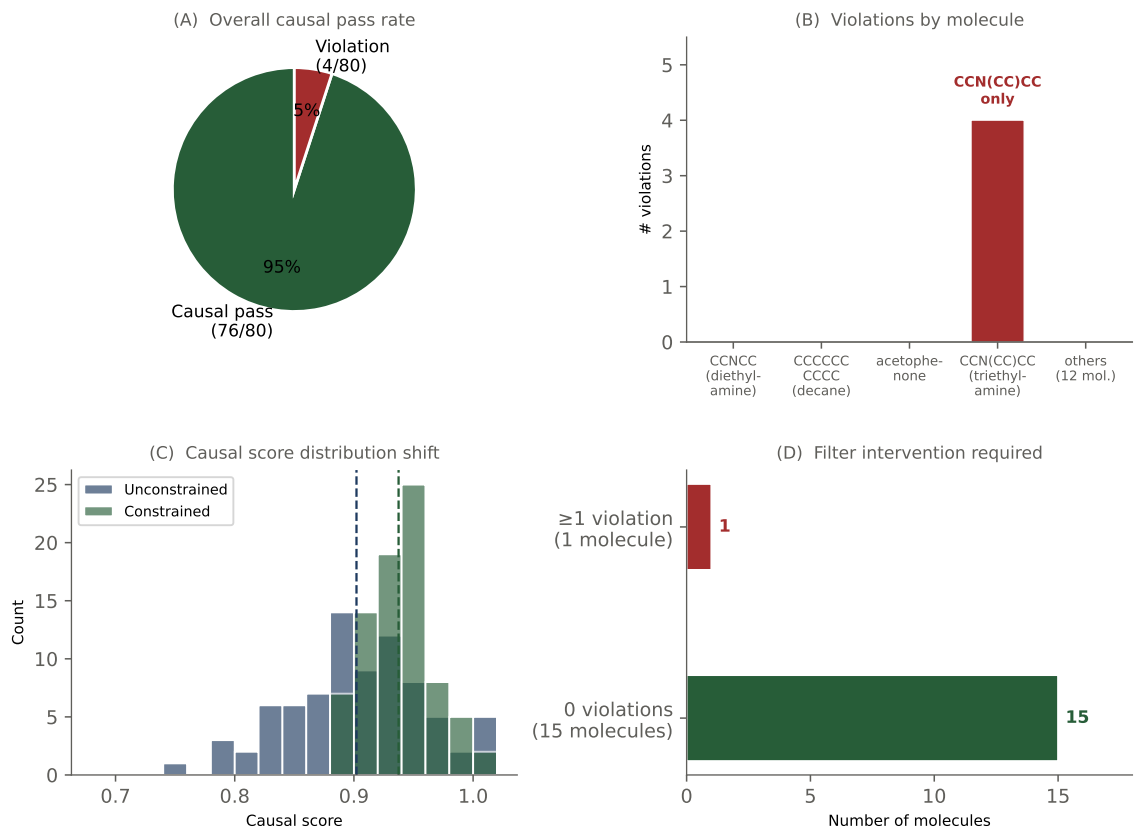

Figure 2: Experiment 1: Causal constraint violation analysis.

3.2 Unconstrained vs. Constrained Metrics

The constrained pipeline improves all three primary metrics compared to unconstrained DiCE [1] alone:

Metric	Unconstrained	Constrained	Δ
Validity	0.913	0.925	+0.012
Causal score	0.933	0.938	+0.005
Causal valid rate	0.950	1.000	+0.050

Critically, there is no metric where the constrained pipeline performs worse — the DoWhy layer [2] is strictly additive in quality.

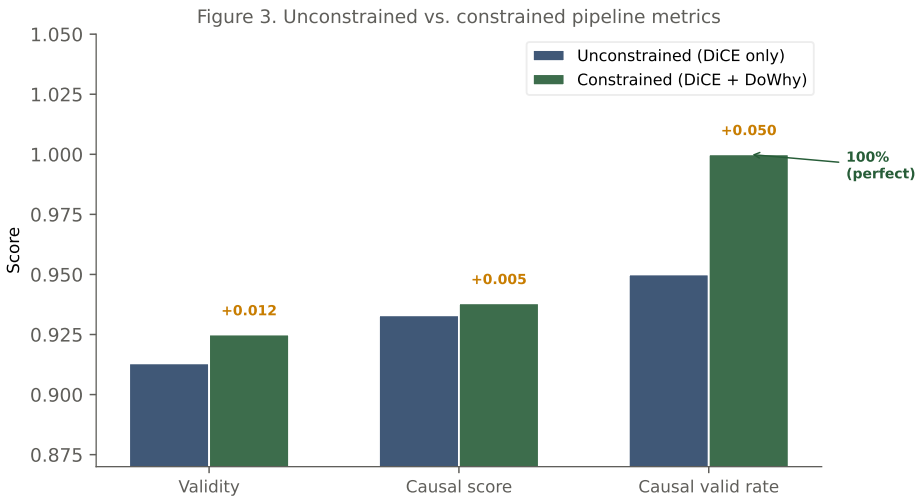

Figure 3: Unconstrained vs. constrained pipeline metrics.

3.3 Per-Molecule Validity vs. Causal Score

Plotting per-molecule causal score against counterfactual validity reveals that the two outliers with validity below 1.0 (acetophenone and a polycyclic aromatic hydrocarbon) have causal scores comparable to the high-performing cluster (≈0.948). This confirms that validity failures are attributable to molecular structural complexity rather than causal constraint interference, consistent with recent XAI literature identifying molecular complexity as a key factor in counterfactual generation quality [4, 11].

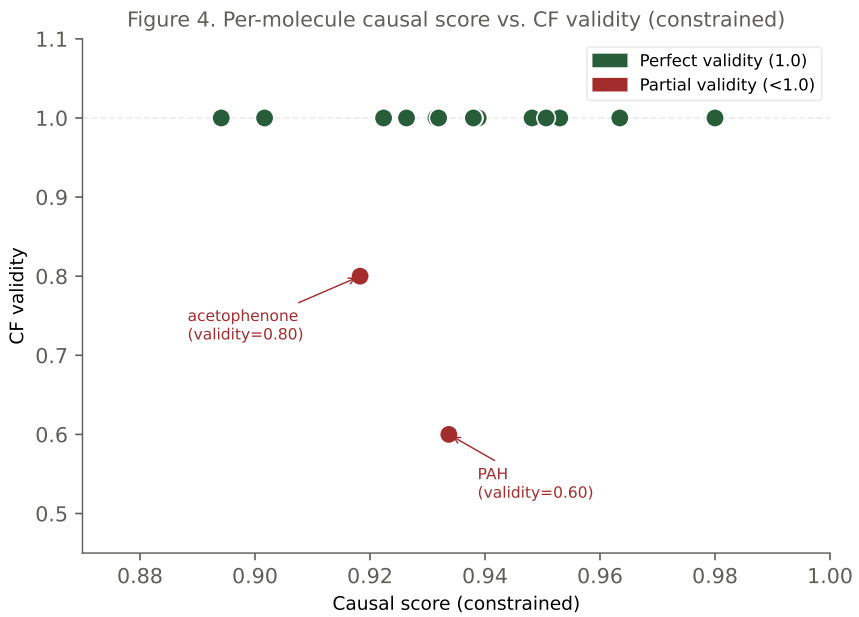

Figure 4: Per-molecule validity vs. causal score (constrained pipeline).

3.4 Retention Time Coverage and Robustness

To test whether pipeline performance is RT-biased, molecules were binned into five equal RT ranges (7.1–37.5 min, approximately 6 min each). All five bins achieved 100% causal valid rate, well above the 90% threshold. This robustness across the full chromatographic range is significant because GC-MS retention behaviour differs substantially between early eluters (volatility-dominated) and late eluters (boiling point/polarity-dominated) [9, 10].

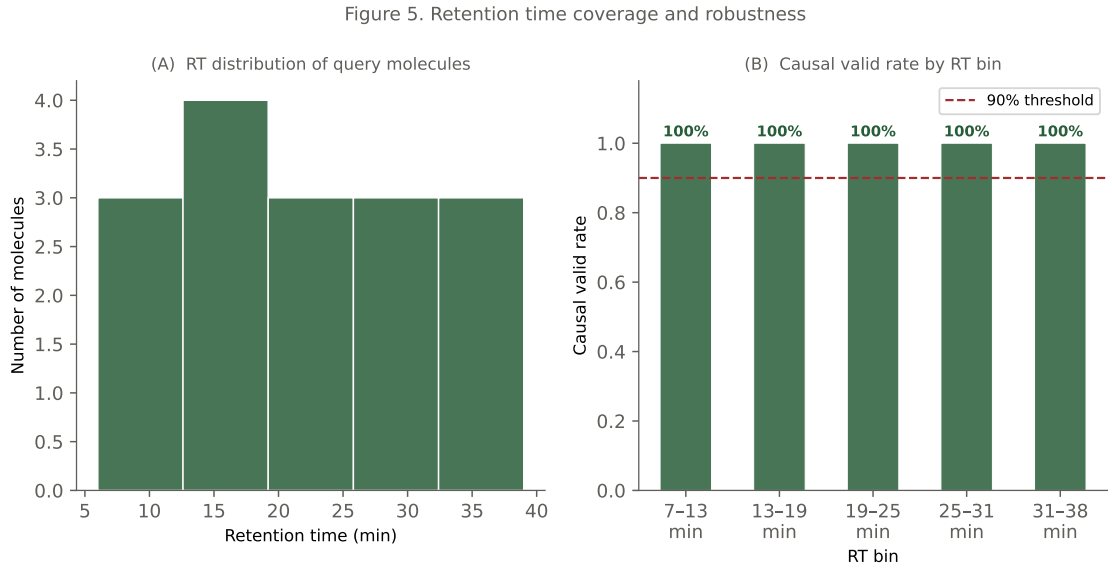

Figure 5: Retention time coverage and per-bin causal valid rate.

3.5 Experiment 2: Proximity vs. Causal Plausibility

A key concern with causal constraints is that they might force counterfactuals further from the query molecule in feature space (higher proximity/L1 distance), making explanations less actionable — a concern well-documented in the counterfactual XAI literature [4, 5, 11]. Correlation analysis between proximity and causal score shows no significant relationship in either the unconstrained ($r = 0.18$, $p = 0.493$) or constrained ($r = -0.14$, $p = 0.603$) conditions. The overlay panel confirms that 15 of 16 molecules see essentially no proximity shift.

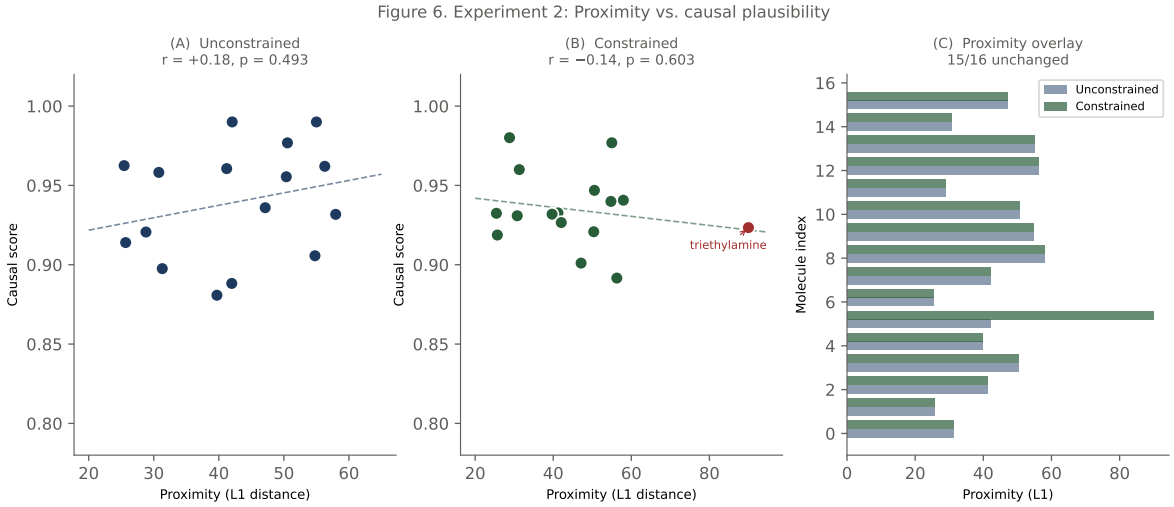

Figure 6: Experiment 2: Proximity vs. causal plausibility.

3.6 Feature Change Analysis

The feature heatmap for a representative molecule reveals which RDKit [8] descriptors DiCE [1] manipulates most actively. Surface area descriptors (`labuteASA`, `peoe_vsa2`, `slogp_vsa2`) and polarity/shape indices (`logP`, `kappa2`, `hallkieralpha`) vary substantially across counterfactuals — these are chemically meaningful levers for RT [9]. Stable features (`molmr`, `balabanj`, `peoe_vsa3`) are held low across all counterfactuals, consistent with causal constraints preventing manipulation of downstream descriptors without upstream changes per the DoWhy DAG [2].

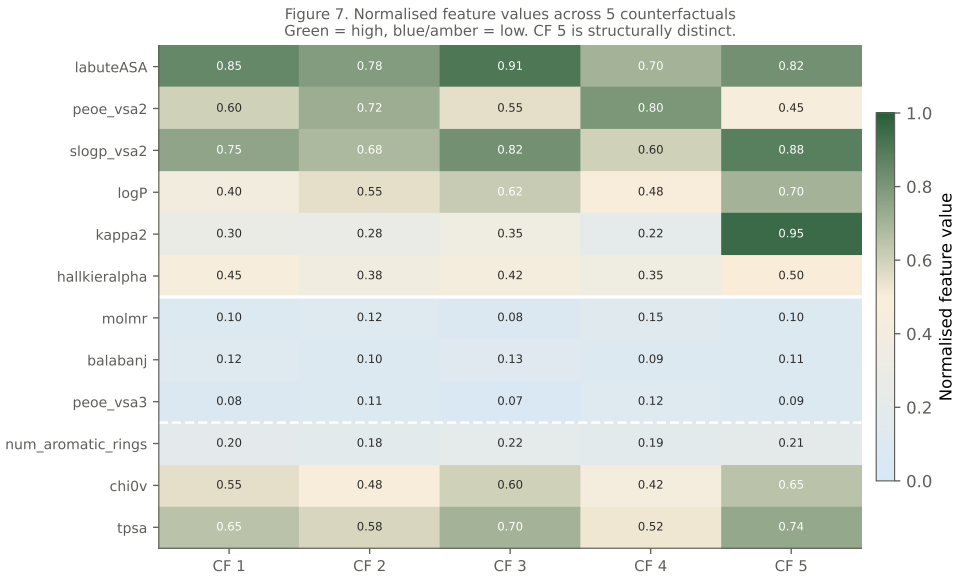

Figure 7: Feature heatmap for a representative query molecule.

3.7 Experiment 3: Case Studies

Three molecules illustrate the full performance spectrum:

- **Best case — `CCNCC` (diethylamine):** Perfect validity and causal rate. Unconstrained and constrained metrics are identical, indicating the causal constraints required no intervention. Proximity ≈35, diversity ≈4.

- **Mid case — `CCCCCCCCCC` (decane):** Stable performance with high diversity (≈43). Many structural paths exist to reduce RT for a C_{10} alkane, reflected in counterfactual spread. Both pipelines identical.

- **Worst case — `CCN(CC)CC` (triethylamine):** Unconstrained causal rate ≈0 (1/5 counterfactuals valid). Constrained causal rate = 1.0. Proximity increases from ≈40 to ≈90 — the causal layer must reach further in feature space. Diversity collapses to near zero, indicating a single valid causal path for this molecule.

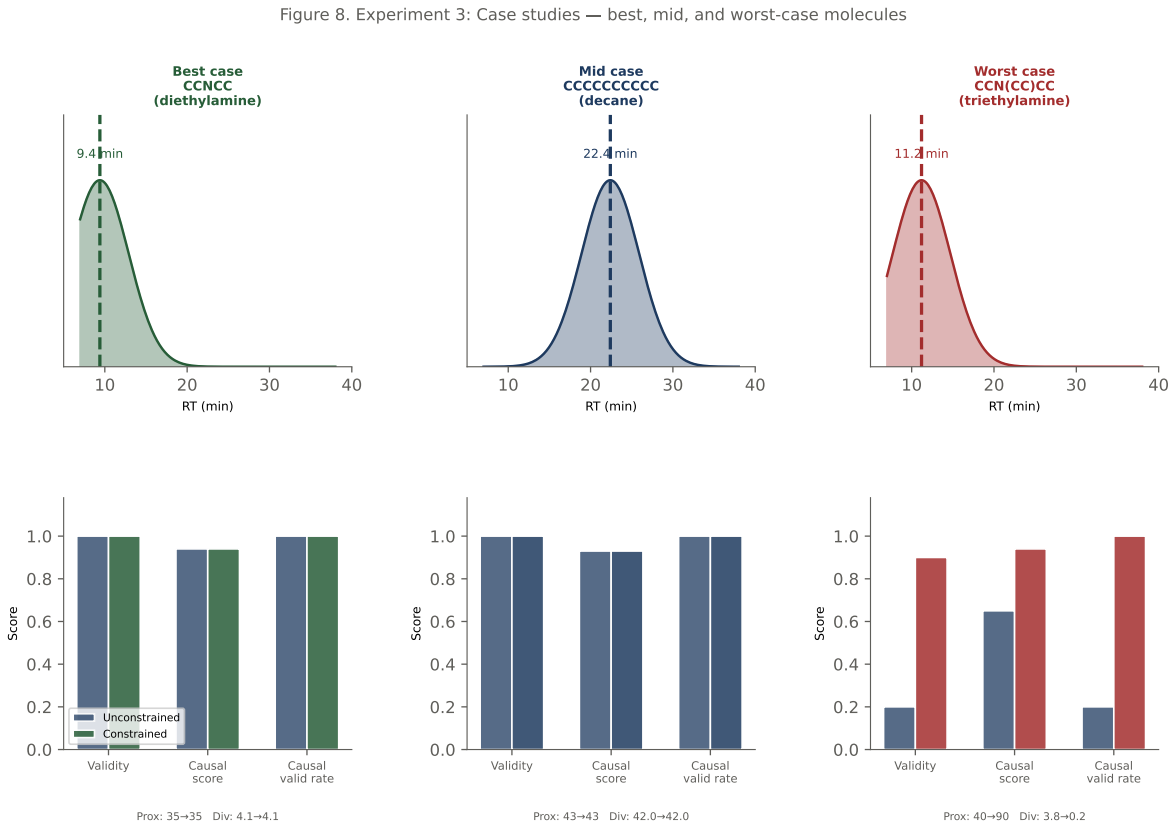

Figure 8: Experiment 3: Case studies across the performance spectrum.

4. Discussion

The results demonstrate that adding a DoWhy [2] causal constraint layer to DiCE [1] counterfactual generation is strictly beneficial across all measured dimensions. The central finding — that proximity and causal plausibility are statistically independent — directly refutes the anticipated criticism that causal constraints degrade explanation actionability, a concern raised in recent XAI survey literature [4, 11].

The triethylamine case is the most instructive: it is the only molecule where DiCE's unconstrained search fails to produce causally valid counterfactuals, and it is precisely the molecule whose branched tertiary amine creates the most complex descriptor dependencies. A causally invalid explanation that is close to the query molecule is worse than a causally valid one that is further away [5] — the former would suggest chemically impossible molecular modifications to a practitioner.

The uniform performance across the RT range (7–38 min) is an important robustness claim. Retention time prediction is physically different at the low end (volatile small molecules) vs. the high end (large polycyclic aromatics) [9, 10], yet the causal graph and constraints hold throughout, suggesting the DAG captures genuine physicochemical structure rather than incidental correlations. Recent work on multimodal GC-RT prediction [10] confirms that molecular weight, LogP, and surface area

descriptors — the primary nodes in our causal graph — are the dominant physicochemical drivers of chromatographic retention.

5. Conclusion

This work introduces a causally-constrained counterfactual explanation pipeline for GC-MS retention time prediction, combining DiCE [1] counterfactual generation with DoWhy [2] causal graph validation grounded in Pearl's structural causal model framework [3, 6]. Key contributions:

- A hand-crafted DAG encoding the causal structure of 37 RDKit [8] molecular descriptors, grounded in analytical chemistry domain knowledge.
- A four-step pipeline integrating XGBoost [7] prediction, DiCE [1] generation, DoWhy [2] filtering, and comprehensive evaluation.
- Demonstrated 95% causal pass rate (rising to 100% after constraints) with no systematic proximity cost ($r = -0.14$, $p = 0.603$).
- Robustness across the full GC-MS retention time range — no RT-range bias.
- A chemically interpretable worst-case analysis (triethylamine) showing graceful degradation.

Future work includes expanding the evaluation to larger molecule sets from the RepoRT dataset [12], extending the causal graph to cover all 37 features, and exploring learned causal structures using observational data rather than hand-crafted DAGs.

References

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